

Handover Deployment Report — UGM Impact Analytics

Date: 2026-08-28 Prepared by: Hermes Agent / BTD Project repo: `dedieko-priyadi/ugm-analytics` Live URL: `https://dts-lab.web.id/ugm-analytics/` Server profile: Docker on NUC, exposed through Cloudflare Tunnel subpath `/ugm-analytics/`

1. Executive Summary

UGM Impact Analytics is live and publicly reachable at `https://dts-lab.web.id/ugm-analytics/`. The deployed application currently serves the `berita-dampak/` Streamlit dashboard successfully. Public rendering, styling, filters, charts, and data reads are working.

However, this handover also records one important server-side code gap:

- the **weekly update pipeline cannot run inside the Linux container as written**, because part of the code still assumes a **Windows virtualenv path** (`venv/Scripts/python.exe`).

So the current state is:

- **dashboard render and public access:** PASS
- **server-side update button / update runner inside container:** FAIL
- **project scope as a whole:** PARTIAL — only the `berita-dampak/` track is operational; other planned subprojects still depend on upstream data access.

2. Scope of This Handover

This handover covers:

- the currently deployed `berita-dampak/` application
- the deployment architecture used on the BTD server
- the verified operational state from the user path
- deployment lessons learned
- developer notes on what should be changed in source code to make the full workflow server-safe

This handover does **not** claim that the following subprojects are operational:

- `kkn-desa-binaan/`
- `mahasiswa-afirmasi/`

Those still require data/access prerequisites outside the deployment layer.

3. Deployment Architecture

Public user

```
-> https://dts-lab.web.id/ugm-analytics/  
-> Cloudflare Tunnel ingress rule (/ugm-analytics/*)  
-> localhost:8766 on NUC  
-> Docker container: ugm-analytics
```

```
-> Streamlit app: berita-dampak/dashboard_berita_dampak.py
-> DuckDB file mount: berita-dampak/data/ugm_news.duckdb
```

Observed deployment configuration:

- container image base: `python:3.10-slim`
- app port: `8766`
- host bind: `127.0.0.1:8766:8766`
- Streamlit base URL path: `/ugm-analytics/`
- Docker subnet: `172.16.93.0/24`
- data mount: `./berita-dampak/data:/app/berita-dampak/data`

4. What Was Verified Live

4.1 Public reachability

Verified from the user path:

- `https://dts-lab.web.id/ugm-analytics/` returns `200`
- final public path remains `/ugm-analytics/`

4.2 Browser render verification

Verified via Playwright headless:

- title: `Analisis Dampak Berita UGM`
- CSS rules loaded: `1199`
- JavaScript runtime errors: `0`
- body text length: `7546`

The deployed UI clearly renders real application content, not a blank Streamlit shell.

4.3 Functional dashboard content present

The following elements were observed in the rendered UI:

- year filter range (`2005` – `2026`)
- 13 impact topic filters
- data source filters (`sitemap` , `rss`)
- Kepmen pillar filters (`Lingkungan` , `Ekonomi` , `Sosial`)
- metric card `Total berita (filter)`
- chart area `Cakupan vs Total Berita UGM per Tahun`
- data-backed aggregate values visible in the page

4.4 Current data state visible

Observed directly in the live UI:

- update timestamp visible in sidebar
- total filtered news visible
- 13 selected impact topics visible
- chart sections render without JS failure

5. Deployment Process Narrative

Step A — repository and deployment artifacts

The repository already contained deployment artifacts for Docker + Cloudflare subpath exposure. Existing reports showed the intended architecture and previous deployment success claims.

Step B — public-state cross-check

A lightweight HTTP probe initially made the app look generic because the first HTML probe surfaced only a generic Streamlit title path. That was **not sufficient evidence** about the actual rendered UI.

Step C — full browser verification

A Playwright render check was then used to verify the real user-facing state. This confirmed that the deployed app is not just reachable — it is actually rendering the expected analytics dashboard with data, controls, and charts.

Step D — source/deployment consistency review

After the app render was validated, the deployment code and runtime assumptions were reviewed against the containerized Linux server environment. That review found a server-side execution bug in the update pipeline path handling.

6. Important Findings

6.1 The deployed app is operational

The current public deployment works for the `berita-dampak/` dashboard itself. No immediate code change is required for the dashboard to render publicly.

6.2 The update workflow is not server-safe yet

This is the main code-level issue that should be handed back to the developer.

Observed in source:

- `berita-dampak/dashboard_berita_dampak.py`
- `berita-dampak/scripts/update_mingguan.py`

Both construct a Python executable path using a Windows-only convention:

```
Path(...)/"venv"/"Scripts"/"python.exe"
```

Observed runtime evidence inside the deployed Linux container:

- `/app/venv/Scripts/python.exe` does **not** exist

Direct server verification:

```
FileNotFoundError: [Errno 2] No such file or directory: '/app/venv/Scripts/python.exe'
```

This means:

- the dashboard page can render
- the **background update button / pipeline trigger path cannot run successfully in the server container** as written

6.3 Project documentation is stale in places

There is an internal documentation mismatch:

- deployment reports say the service is live and verified
- `deploy/README.md` still says the deploy target has not been executed
- project registry metadata also appears stale relative to the actual runtime state

This is not a runtime blocker, but it is a handover-quality issue.

6.4 Full project scope is broader than the live app

The live deployment currently represents the `berita-dampak/` subproject only. Planned scope still includes:

- `matkul-sustainability/` (finished as analysis work, but not verified as part of the live public endpoint in this review)
- `kkn-desa-binaan/` (blocked by eLOK access)
- `mahasiswa-afirmasi/` (blocked by official/sensitive data access)

So if the developer expects the public URL to represent the full multi-track UGM Analytics program, additional product work is still needed.

7. Developer Notes — Changes Recommended for Server Compatibility

Change 1 — Remove Windows-specific Python path assumptions

Current problem

The code assumes the Python interpreter is located at:

```
venv/Scripts/python.exe
```

That is valid on Windows, but not inside the Linux Docker container.

Recommended fix

Use one of these patterns instead:

Option A — use the current interpreter

```
import sys
PY = sys.executable
```

This is the best default for subprocess calls that should run in the same environment as the app.

Option B — use an environment variable override

```
import os, sys
PY = os.environ.get("UGM_ANALYTICS_PYTHON", sys.executable)
```

This allows the deployment environment to override the interpreter path explicitly when needed.

Applies to

- `berita-dampak/scripts/update_mingguan.py`
- `berita-dampak/dashboard_berita_dampak.py` (update button subprocess launcher)
- any other script still documented or coded with `venv/Scripts/python.exe`

Change 2 — Keep subprocess invocation OS-agnostic

The code currently uses:

```
creationflags=getattr(subprocess, "CREATE_NO_WINDOW", 0)
```

This is acceptable because it gracefully falls back to `0` on Linux. No urgent fix is required here.

However, the subprocess launcher should still be reviewed together with interpreter selection so the whole background update flow is platform-neutral.

Change 3 — Consider separating server update execution from UI button execution

The current app design allows the dashboard UI to trigger a full update pipeline. That is convenient for local/operator use, but for server deployments it may be better to run updates as:

- a cron/scheduler job
- a supervised background task
- or a dedicated admin-only action outside the public dashboard path

Why this matters:

- long-running updates can lock the DuckDB file
- dashboard refresh during update can degrade operator experience
- server runtime is more predictable when data refresh is scheduled out-of-band

This is a design recommendation, not a deployment blocker.

Change 4 — Refresh repository docs so they match runtime truth

Recommended documentation cleanup:

- update `deploy/README.md` to say the deployment is already live
- update registry progress metadata to reflect real deployment state
- keep one authoritative handover/deployment report for future operators

8. What Does NOT Need Immediate Code Change

The following are already working in the deployed app:

- public subpath exposure through Cloudflare Tunnel
- Streamlit `baseUrlPath=/ugm-analytics/`
- public rendering and CSS/JS load
- DuckDB live file mount for dashboard reads
- current chart/filter rendering behavior

So the main required code fix is **not** the public deployment shell. It is the **server-side update execution path**.

9. Operational Recommendation

Short-term operator recommendation:

- keep the public dashboard live as-is
- do **not** rely on the in-dashboard update trigger until interpreter-path handling is fixed
- run updates via a controlled server-side command path instead

Developer recommendation:

- patch the interpreter path logic first
- then re-test the update button / update runner in the Linux container
- after that, decide whether updates stay UI-triggered or move to scheduled execution

10. Handover Conclusion

UGM Impact Analytics is successfully deployed and publicly accessible for the `berita-dampak/` Streamlit dashboard. The dashboard itself renders correctly and reads data successfully from the live DuckDB mount.

The main code change still recommended for the developer is:

- replace Windows-only interpreter assumptions with server-safe interpreter resolution

Without that patch, the public dashboard can be viewed, but the built-in update workflow cannot reliably run inside the deployed Linux container.

Prepared from live verification, repository inspection, and container-side execution evidence on 2026-08-28.